

JavaScript DSA Complete Guide

This PDF contains a complete JavaScript Data Structures and Algorithms guide.

It includes:

- Introduction to DSA and Big-O Notation
- All important Data Structures explained with operations and JS code
- 300 practice problems arranged by difficulty:
Beginner (100), Intermediate (100), Master (100)
- Each problem includes statement, approach, JS solution, and complexity analysis

Section 1: Data Structures

1. Arrays

Definition: Arrays are ordered collections of elements that can be accessed by index.

Common Operations:

- Traversal
- Insertion
- Deletion
- Searching
- Sorting

JavaScript Example:

```
let arr = [1, 2, 3, 4];  
arr.push(5); // Insert at end  
arr.splice(2, 0, 10); // Insert 10 at index 2  
console.log(arr); // [1, 2, 10, 3, 4, 5]
```

Section 2: Beginner Problems

Problem 1: Find Maximum in Array

Statement: Given an array of integers, find the maximum element.

Approach: Traverse the array, keep track of the maximum found so far.

JavaScript Solution:

```
function findMax(arr) {  
    let max = arr[0];  
    for(let i=1; i<arr.length; i++) {  
        if(arr[i] > max) max = arr[i];  
    }  
    return max;  
}  
console.log(findMax([1, 5, 3, 9, 2])); // 9
```

Time Complexity: $O(n)$

Space Complexity: $O(1)$